Horticulture 1 · Unit 7 · Level 2 (modified)

Propagation



TODAY'S PLAN

- 🌱 Sexual vs. asexual
- 🎋 Stem and leaf cuttings
- 🐣 Separation and division
- 🥔 Vegetable propagation

Review · 5 minutes

Quick Review



- 🌰 Name the 3 parts of a seed.
- 🌱 Cotyledons vs. true leaves?
- 🏡 What is indirect seeding?

Words to Know

Words to Know



Propagation

Making new plants, either from seed (sexual) or from plant parts (asexual).



Sexual propagation

Growing plants from seeds. The new plant is a mix of two parents.



Asexual propagation

Growing plants from cuttings, division or runners. The new plant is a copy (clone) of the parent.



Stem cutting

A piece of stem with leaves, placed in mix so it grows roots.

Words to Know

More Words to Know



🌿 **Leaf cutting**: A leaf or part of a leaf that grows roots and a new plant (succulents, African violets).

🐣 **Separation**: Pulling off small offsets ('chicks') from a parent plant and potting them.

✋ **Division**: Cutting or pulling a clump (like an iris or hosta) into pieces with roots.

🧪 **Rooting hormone**: A powder or gel that helps cuttings grow roots faster. Use gloves.

Compare

Sexual vs. Asexual



🌰

Sexual (seed)

Two parents
Offspring vary

✂️

Asexual (parts)

One parent
Clone: exact copy

🃏 Check 1 of 3



Why do growers clone a plant?

Hold up 1, 2, or 3:

1️⃣ To change its color

2️⃣ To get exact copies of a plant they like

3️⃣ To make seeds

✅ Answer · Card 1 of 3



To get exact copies of a plant they like

Same flower, color and habit.

Steps

Stem Cutting



1. With an adult, cut 3-4 inches below a node
2. Remove the lower leaves
3. Dip the end in rooting hormone (gloves)
4. Stick in moist mix; keep humid
5. Roots in 2-4 weeks

Learn

Leaf Cuttings



Succulents, African violets and snake plants grow from **leaves**.

Let succulent leaves dry (callus) a few days, then lay on mix.

New roots and a tiny plant form at the base.

Learn

Separation and Division



Separation: pull off natural offsets, like hen-and-chicks pups.

Division: split a clump (iris, hosta) into pieces, each with roots and shoots.

Fast and easy for perennials.

🃏 Check 2 of 3



Splitting an iris clump into pieces is…

Hold up 1, 2, or 3:

1️⃣ Seeding

2️⃣ Grafting

3️⃣ Division

✅ Answer · Card 2 of 3



Division

Each piece has roots.

Learn

Vegetable Propagation



Potatoes
seed pieces with eyes



Strawberries
runners

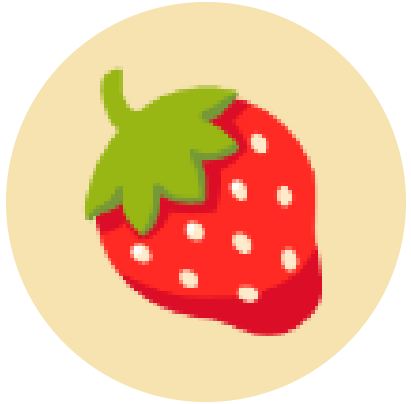


Garlic
cloves



Sweet potatoes
slips

🃏 Check 3 of 3



Strawberries make new plants from…

Hold up 1, 2, or 3:

1️⃣ Runners

2️⃣ Cloves

3️⃣ Pine cones

✅ Answer · Card 3 of 3



Runners

Runners root at the tip.

Learn

Rooting Hormone Safety



Wear **gloves**; pour a little into a cup, never dip into the jar.

Don't touch your face; never ingest.

Wash hands when done.

Wrap-Up

Recap



Sexual
seeds, variety



Asexual
clone



Stem cutting
below a node



Division
split clumps

Wrap-Up

Exit Ticket



Sexual vs. asexual: one difference.

List the steps of a stem cutting.

Name 2 vegetables propagated asexually and how.